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Question Paper Code : 19935

B.E. / B.TECH. DEGREE EXAMINATIONS, MAY / JUNE 2023

Fourth Semester

B.Tech. – Artificial Intelligence and Data Science

19AD403 - PROGRAMMING WITH JAVA

(Regulations: Mepco – R2019)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

BTL, CO

PART A – (10 × 2 = 20 Marks)

U, CO1 1. Illustrate the access condition when there is no access specifier in the declaration of a class.

U, CO1 2. What will be the output of the following program? Justify your answer.

```
class X{
X(){
System.out.println("Constructor of X is called");
} }
class Y extends X{
Y(){
System.out.println("Constructor of Y is called");
} }
class Z extends Y{
Z(){
System.out.println("Constructor of Z is called");
} }
class Main{
public static void main(String[] args){
        Y objZ=new Z();
        }
}
```

A) Constructor of Class Z only will be invoked

B) Constructor of Class Y and Z constructor only will be invoked

C) Constructor of Class Z and all their ancestor's constructor will be invoked

D) Constructor of Class Y only will be invoked

U, CO2 3. Differentiate between checked and unchecked Exception with a suitable example.

U, CO2 4. Which of the following interface must contain a unique element? Justify your answer.

A) List B) Array C) Set D) Collection

A, CO3 5. FileInputStream and FileOutputStream classes are recommended for character input from file and output to file respectively. (True/False). Justify your answer.

U, CO3 6. Write short notes on java.io.Serializable.

U, CO4 7. An Applet can be embedded in which of the following? Give example for the scenario. Justify your answer.

A) RTF File B) html file C) MS Word D) gif file

U, CO4 8. List the types of containers in Java AWT. Give example code for any one container.

U, CO5 9. How will you wake-up a blocking or waiting thread?

U, CO5 10. What is the output of the following code? Justify your answer.

```
public class Main{
public static void main(String[] args) {
Thread t = Thread.currentThread();
t.setName("SUCCESS");
System.out.println(t);
 }
}
```

A) Thread[5,main] B) Thread[SUCCESS,5]

C) Thread[main,5,main] D) Thread[SUCCESS,5,main]

PART B – (5 × 16 = 80 Marks)

A,CO1 11. a) i. In a certain school, the health check-up of students of a particular class was conducted. The following table was prepared showing the record of a group of 10 students:

Student Name	Weight	Height
Student 1		
Student 2		
Student 3		
..		
..		
Student 10		

Write a Java Program using 2D array to compute the following:

A) Retrieve the students those who having highest and lowest value of height respectively.

B) Retrieve the students those who having highest and lowest value of weight respectively. (10 Marks)

U,CO1 11. a) ii. Write short notes on Inheritance and its types in Java. (6 Marks)

OR

A,CO1 11. b) i. Write Program in which three interfaces are defined, namely Inf1, Inf2 and Inf3. Inf2 extends Inf1; Inf3 extends Inf2 and Inf1.

Inf1 declares method withdraw()

Inf2 declares method deposit()

Inf3 declares method display() that shows the balance account number

A) Create a class (accountNo, name, address, balance)

B) Implement the aforementioned three interfaces and accessing it. (10 Marks)

U,CO1 11. b) ii. Explain about the Cosmic super class. (6 Marks)

A,CO2 12. a) i. Develop a java code to create custom exception class for the output when a negative number is placed as argument for sqrt() method of Math class. (10 Marks)

U,CO2 12. a) ii. Illustrate different methods of dereferencing the objects using Garbage collection. (6 Marks)

OR

A,CO2 12. b) i. Create a class called Customer with attributes(Id, Name, address) and implement the following:

A) Create N instances of Customer class

B) Add display N instances toTreeSet collection and display it. (10 Marks)

U,CO2 12. b) ii. Differentiate Set and List Interface with a suitable example. (6 Marks)

A,CO3 13. a) i. Illustrate Java code for sorting N strings and perform the following:

A) Write the strings into file called ‘Result.txt’

B) Read the strings from the file ‘Result.txt’ and display it into user (10 Marks)

U,CO3 13. a) ii. Write short notes on standard IO streams of Java. (6 Marks)

OR

A,CO3	13. b) i.	Design a class called Points with the data members like x_val and y_val. Include the constructors for initializing the data members and a method to display the Points details. Create two Point objects and write the objects into a File. Also, read the content of the File and display the Point details.	(10 Marks)
U,CO3	13. b) ii.	How can you handle byte oriented data in Java?	(6 Marks)
A,CO4	14. a) i.	Design a GUI Application for creating Three Labels called "One", "Two", "Three" which follows Flow Layout along with Label texted as "Press button of your choice". After clicking any one of button, the label name get changed to the respective button name.	(10 Marks)
U,CO4	14. a) ii.	Identify and explain appropriate Even Handling mechanism along with their interface for the person access the application through the input which is made by a keyboard.	(6 Marks)
		OR	
A,CO4	14. b) i.	Design an Applet Window with suitable AWT text fields along with labels such as Name, Age and Weight. If the user presses "ENTER" key after filling the text fields, the information is passed on to another set of labels for the respective text field.	(10 Marks)
U,CO4	14. b) ii.	Assume that there is Online Shopping cart- "ABC-Cart" which can handle the more number of transactions on a day. Three-tier architecture must be followed for its implementation. Elucidate the necessary steps for setting up JDBC environment for the above application.	(6 Marks)
U,CO5	15. a) i.	Explain the different methods to create new Thread with suitable examples.	(10 Marks)
A,CO5	15. a) ii.	Write a program that creates a thread in addition to the main thread and runs the code for finding sum of cube of individual digits.	(6 Marks)
		OR	
U,CO5	15. b) i.	Explain in detail about Thread Synchronization and Race Condition.	(10 Marks)
A,CO5	15. b) ii.	Write a program to illustrate the thread priority for the thread 'First' and 'Second'.	(6 Marks)